

Saptarshi Mandal

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Education

University of Illinois Urbana-Champaign

PhD in ECE(Ongoing) ; Advisor - R. Srikant

GPA : 4/4

Aug 2021 - Dec 2026

Indian Institute of Technology, Kharagpur

Joint B.Tech-M.Tech, major in EECE and minor in CSE

GPA : 9.34/10

Aug 2016 - April 2021

Internships

Google Research

Supervisor - Prof. Milind Tambe, Harvard University; Collaborator: Assistant Prof. Niclas Boehmer, HPI, Germany

AI for Social Good

Oct - Dec, 2023

- Explored the use of data-driven approaches to efficiently allocate scarce resources and encourage patients to seek healthcare in rural India. Contributed to the modeling of general Restless Bandit framework suitable for the intervention effect dataset in Google Research, India .
- Accepted as a full paper at **International Conference on Autonomous Agents and Multiagent Systems (AAMAS) 2025**.
- Link to the paper: <https://arxiv.org/abs/2501.06103>.

NVIDIA

Project intern-digital profile

Bangalore, India

Summer, 2019

- Automated the fault aggregation part of error collation from the IP metadata of the Orin SoC architecture accessing the Google Drive API in the Pearl script and parsing IP metadata efficiently.
- Built a comprehensive latency calculator using the maps between IPs in the same SoC architecture. Exponentially reduced the latency calculation time using shortest path algorithms and recursion in Python.

Research

A Theoretical Analysis of Soft-Label vs Hard-Label Training in Neural Networks

Supervisors - : Prof. Lin Xiaojun, CUHK; Prof. R. Srikant, UIUC

UIUC

2023 - 2024

- Theoretical insights for a few widely known phenomena of Knowledge Distillation including Model Compression (using traditional labels vs. soft labels for training in Supervised Classification task) using the tools of Neural Tangent Kernel.
- Accepted at the **Learning for Decision and Control (L4DC) 2025**.
- Link to the paper : <https://arxiv.org/abs/2412.09579>.

Inferring Labels from Multi-type Crowdsourced Data

Supervisor - Prof. R. Srikant, UIUC; Collaborators: Seo Taek Kong, Dimitrios Katselis

UIUC

2021 - 2022

- Extended the Dawid-Skene (DS) model to multi-type cases and analyzed the error bounds of existing algorithms. Focusing on the case where there are two types of tasks, we propose a spectral method to partition tasks into two groups that cluster tasks by type. Our analysis reveals that task types can be perfectly recovered if the number of workers scales logarithmically with the number of tasks.
- Accepted for a talk at **INFORMS Applied Probability Society Conference**.
- Under Review **Transaction on Machine Learning Research (TMLR)**.
- Link to the Paper: <https://arxiv.org/abs/2302.07393>.

An Online Delay-reliable Efficient Algorithm for Multi-resource Allocation

Supervisor - Prof. Goutam Das, IIT Kharagpur

IIT Kharagpur

2020 - 2021

- Proposed a polynomial time delay reliable scheduling strategy for multi-resource allocation in wired media optimal in the presence of adversarial packet arrival; Proposed an approximate solution for multiple types of packets for the same.
- Under preparation for GLOBECOM 2025.

Parameter Calibration for EESM and TIESNR for 5G NR

Supervisor - Prof. Neelesh B. Mehta, IISc

Indian Institute of Science (IISc)

2020

- Calibrated the parameters of two widely used Effective SNR mappings(link quality matrices) - EESM(Exponential Effective SNR mapping) and TIESNR(Time-invariant Effective SNR) in 5G radio through link level simulations in MATLAB 5G Toolbox.

Relevant Graduate Courses with Grades

Machine Learning	Stat. Learning Theory(A), Stat. Reinforcement Learning(A), Neural Networks and Applications(A)
Probability	Random Processes(A+), High Dimensional Prob.(A+), Information Theory (A+), Queueing Theory(A+)
Optimization, Game Theory	Adaptive Systems(A+), Network Optimization(A+), Game Theory(A+)
Algorithms	Approximation and Online Algorithms(A), Communication Signal Processing and Algorithms(A+)

Teaching Experience

UIUC	Graduate Course: “Introduction to Optimization”	2024
IIT Kharagpur	Under-Graduate Lab: “VLSI Lab”	2021
IIT Kharagpur	Under-Graduate Lab: “Microcontroller Systems Lab”	2020

Technical Skills

Programming Languages	Python, PyTorch, Matlab, C, C++, verilog
Software and Platforms	Jupyter Notebook, Simulink, Omnetpp, CPLEX, Xilinx ISE, $\LaTeX$, Beamer, git
Languages	English(Fluent), Hindi(Fluent), Bengali(Native)

Awards and Honors

IISc	Scholarship: “KVPY ”	2016
IISc	Fellowship: “Summer Research Fellow” by “Indian Academy of Science”	2020

Extra Academic Activity

- Former Hardware Modelling Team Member, LBS hall, IIT Kharagpur
- Former captain, Bengali Dramatics - Nehru Hall, IIT Kharagpur
- A Master of Ceremonies in the main event in Bengali Student Organization, UIUC